

GENERALIZATIONS TO NON-NOETHERIAN RINGS

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In the following, we extend some theorems from [1] to include the case where the ground ring is non-noetherian. In Corollary 1.11 we show that an open immersion is an algebra of finite presentation, which was stated without proof in [1, Example 9.2.4 (3)]. Lemma 1.12 is a local to global lemma for a homomorphism of finite presentation which generalizes [1, Lemma 2.1.5]. In Lemma 1.5 we prove [1, Example 9.2.2 (4) and (5)]. Theorem 2.1, a generalization of [1, Theorem 9.3.9], proves that locally an étale algebra is isomorphic to a standard étale algebra. Theorem 2.3, a generalization of [1, Proposition 9.2.9], proves that if S is an étale R -algebra, then $\text{Spec } S \rightarrow \text{Spec } R$ is an open map. Theorem 3.6, which generalizes [1, Theorem 9.3.12], is a theorem of permanence for an étale algebra over a non-noetherian ground ring. The proofs below utilize only theorems proven in [3] and [1].

1. ALGEBRAS OF FINITE PRESENTATION

One goal of this section is to prove that if $f : R \rightarrow S$ is a homomorphism of commutative rings and $f^\# : \text{Spec } S \rightarrow \text{Spec } R$ is an open immersion, then S is an R -algebra of finite presentation (see Corollary 1.11 below). This theorem is referenced without proof in [1, Example 9.2.4 (3)]. The corollary and its proof presented here are based on [4, Proposition I.4.6, p. 328]. We begin by proving some results on algebras of finite presentation.

Throughout, all rings are commutative. Let M be an R -module and α a nonzero element of R . Denote by M_α the localization of M with respect to the multiplicative set $\{1, \alpha, \alpha^2, \alpha^3, \dots\}$.

Definition 1.1. Let R be a commutative ring and M an R -module. We say M is *locally finitely generated*, if there exist elements $\alpha_1, \dots, \alpha_n$ in R such that $R = R\alpha_1 + \dots + R\alpha_n$ and for each i , M_{α_i} is a finitely generated R_{α_i} -module.

Lemma 1.2. *Let M be an R -module. If M is locally finitely generated, then M is finitely generated.*

Proof. Let $\alpha_1, \dots, \alpha_n$ in R such that $R = R\alpha_1 + \dots + R\alpha_n$ and for each i , M_{α_i} is a finitely generated R_{α_i} -module. For each i , let $\{x_{ij} \mid 1 \leq j \leq n_i\}$ be a subset of M which is a generating set for M_{α_i} as an R_{α_i} -module. The reader should verify that $\bigcup_i \{x_{ij} \mid 1 \leq j \leq n_i\}$ is a generating set for the R -module M . $\square$

Definition 1.3. Let S be a commutative R -algebra with structure homomorphism $\theta : R \rightarrow S$. We say S is an *R -algebra of finite presentation*, if there is a polynomial ring in $n \geq 1$ variables over R and a surjective R -algebra homomorphism $\phi : R[x_1, \dots, x_n] \rightarrow S$ such that the kernel of ϕ is a finitely generated ideal. The R -algebra S is said to be *locally of finite presentation*, if there exist elements $\beta_1, \dots, \beta_n$

in S such that $S = S\beta_1 + \cdots + S\beta_n$ and for each i , $S\beta_i$ is an R -algebra of finite presentation. We also say θ is *locally of finite presentation*.

Example 1.4. Let α be a non-zero element of R . Define $\phi : R[x] \rightarrow R_\alpha$ by $\phi(x) = 1/\alpha$. The kernel of ϕ is the principal ideal generated by $x\alpha - 1$. This shows R_α is an R -algebra of finite presentation.

Lemma 1.5. ([1, Example 9.2.2 (4) and (5)])

- (1) (*Change of Base*) Let $f : R \rightarrow T$ be a homomorphism of commutative rings such that T is an R -algebra of finite presentation. If $g : R \rightarrow S$ is any homomorphism of commutative rings, then $f \otimes 1 : S \rightarrow T \otimes_R S$ makes $T \otimes_R S$ into an S -algebra of finite presentation.
- (2) (*Finitely Presented over Finitely Presented is Finitely Presented*) If S is a finitely presented R -algebra and T is a finitely presented S -algebra, then T is a finitely presented R -algebra.

Proof. (1): There exists an onto homomorphism $\phi : R[x_1, \dots, x_n] \rightarrow T$ such that $\ker \phi$ is a finitely generated ideal. Then $\phi \otimes 1 : S[x_1, \dots, x_n] \rightarrow T \otimes_R S$ is onto. The kernel of $\phi \otimes 1$ is generated by the image of $\ker \phi \otimes_R S$, hence is a finitely generated ideal.

(2): There exist homomorphisms $\phi : R[x_1, \dots, x_m] \rightarrow S$ and $\psi : S[y_1, \dots, y_n] \rightarrow T$ such that ϕ and ψ are both onto, and $\ker \phi$ and $\ker \psi$ are both finitely generated ideals. Define $\chi : R[x_1, \dots, x_m][y_1, \dots, y_n] \rightarrow T$ by $\chi(x_i) = \psi\phi(x_i)$ for each i and $\chi(y_j) = \psi(y_j)$ for each j . Then χ is onto and the kernel of χ is generated by $\ker \phi + \ker \psi$, hence is finitely generated. $\square$

Proposition 1.6. Let $f : R \rightarrow S$ be a homomorphism of commutative rings. The following are true.

- (1) If the kernel of f is a finitely generated ideal and f is onto, then S is an R -algebra of finite presentation.
- (2) If f is one-to-one and S is a finitely generated R -module, then S is an R -algebra of finite presentation.
- (3) If the kernel of f is a finitely generated ideal and S is a finitely generated R -module, then S is an R -algebra of finite presentation.

Proof. The proof is left to the reader as an exercise. $\square$

Proposition 1.7. (1) If $f : R \rightarrow S$ is a local isomorphism, then f is locally of finite presentation.

- (2) (*Composition*) If $f : R \rightarrow S$ and $g : S \rightarrow T$ are both locally of finite presentation, then $g \circ f : R \rightarrow T$ is locally of finite presentation.
- (3) (*Change of Base*) Let $f : R \rightarrow T$ and $g : R \rightarrow S$. If f is locally of finite presentation, then $f \otimes 1 : S \rightarrow T \otimes_R S$ makes $T \otimes_R S$ into an S -algebra that is locally of finite presentation.
- (4) (*Tensor Product*) Let $f : S \rightarrow S'$ and $g : T \rightarrow T'$ be R -algebra homomorphisms. If f and g are both locally of finite presentation, then $f \otimes g : S \otimes_R T \rightarrow S' \otimes_R T'$ is locally of finite presentation.
- (5) (*Descent*) If $f : R \rightarrow S$ is locally of finite type and $g : S \rightarrow T$, and $g \circ f : R \rightarrow T$ is locally of finite presentation, then g is locally of finite presentation.

Proof. (1): For every $\mathfrak{q} \in \text{Spec } S$ there exists $\alpha \in R$ such that $f(\alpha) \in S - \mathfrak{q}$ and $R_\alpha \rightarrow S_\alpha$ is an isomorphism. By Example 1.4, S_α is an R -algebra of finite presentation. By [3, Exercise 3.3.29], S is locally of finite presentation.

(2): Let $\mathfrak{r} \in \text{Spec } T$ be arbitrary. There exists $\beta \in T - \mathfrak{r}$ such that $g : S \rightarrow T_\beta$ is of finite presentation. Let $\mathfrak{q} = \mathfrak{r} \cap S$. There exists $\alpha \in S - \mathfrak{q}$ such that $f : R \rightarrow S_\alpha$ is of finite presentation. By Lemma 1.5 (1), $g : S_\alpha \rightarrow T_{\beta\alpha}$ is of finite presentation. By Lemma 1.5 (2), $g \circ f : R \rightarrow S_\alpha \rightarrow T_{\beta\alpha}$ is of finite presentation. Because $\text{Spec } S$ is compact ([3, Exercise 3.3.29]), we see that T is locally of finite presentation as an R -algebra.

(3): Start with elements $\beta_1, \dots, \beta_n$ in T such that $T = T\beta_1 + \dots + T\beta_n$ and for each i , T_{β_i} is an R -algebra of finite presentation. Then $\beta_1 \otimes 1, \dots, \beta_n \otimes 1$ generate the unit ideal in $T \otimes_R S$. By Lemma 1.5 (1), $T_{\beta_i} \otimes_R S$ is an S -algebra of finite presentation, for each i .

(4): By (3), $f \otimes 1 : S \otimes_R T \rightarrow S' \otimes_R T$ and $1 \otimes g : S' \otimes_R T \rightarrow S' \otimes_R T'$ are both locally of finite presentation. By (2), $f \otimes g$ is locally of finite presentation.

(5): First note that $S \otimes_{S \otimes_R S} (T \otimes_R S)$ is isomorphic to T . The diagram

$$\begin{array}{ccc}
 & T & \\
 \rho \nearrow & & \nwarrow g \\
 T \otimes_R S & & S \\
 g \otimes 1 \nwarrow & & \nearrow \mu \\
 & S \otimes_R S &
 \end{array}$$

commutes, where $\rho(t \otimes s) = ts$. By [3, Proposition 3.5.14], f makes S into a finitely generated R -algebra. By [3, Lemma 10.1.6], the multiplication homomorphism $\mu : S \otimes_R S \rightarrow S$ makes S into a finitely presented $S \otimes_R S$ -module. By Proposition 1.6, S is a finitely presented $S \otimes_R S$ -algebra. By (3), ρ makes T into a $T \otimes_R S$ -algebra of finite presentation. By (3) again, $(g \circ f) \otimes 1 : S \rightarrow T \otimes_R S$ is locally of finite presentation. The diagram

$$\begin{array}{ccc}
 S & \xrightarrow{g} & T \\
 (g \circ f) \otimes 1 \searrow & & \nearrow \rho \\
 & T \otimes_R S &
 \end{array}$$

commutes, where $\rho(t \otimes s) = ts$. By (2), g is locally of finite presentation. □

Lemma 1.8. *Let I be a proper ideal in R and α a nonzero element of R such that R_α/I_α is an R -algebra of finite presentation. Then I_α is a finitely generated ideal in R_α .*

Proof. There exists an onto R -algebra homomorphism $\phi : R[x_1, \dots, x_n] \rightarrow R_\alpha/I_\alpha$ such that $\ker \phi$ is a finitely generated ideal. Tensor with $(\cdot) \otimes_R R_\alpha$ to get the exact sequence

$$0 \rightarrow \ker \phi \otimes_R R_\alpha \rightarrow R_\alpha[x_1, \dots, x_n] \xrightarrow{\tau} R_\alpha/I_\alpha \rightarrow 0$$

where $\tau = \phi \otimes 1$. Therefore, $\ker \tau = \ker \phi \otimes_R R_\alpha$ is a finitely generated ideal. Let $\eta : R_\alpha \rightarrow R_\alpha/I_\alpha$ be the natural map. Since τ is an R_α -algebra homomorphism,

there exists a lifting $\psi : R_\alpha[x_1, \dots, x_n] \rightarrow R_\alpha$ such that $\eta\psi(x_i) = \tau(x_i)$ for each i . The diagram

$$\begin{array}{ccccccc} 0 & \longrightarrow & \ker \tau & \longrightarrow & R_\alpha[x_1, \dots, x_n] & \xrightarrow{\tau} & R_\alpha/I_\alpha \longrightarrow 0 \\ & & \downarrow & & \downarrow \psi & & \downarrow u \\ 0 & \longrightarrow & I_\alpha & \longrightarrow & R_\alpha & \xrightarrow{\eta} & R_\alpha/I_\alpha \longrightarrow 0 \end{array}$$

commutes and the rows are exact sequences. Since ψ is onto and u is the identity map, the Snake Lemma ([3, Theorem 2.3.2]) implies $\psi(\ker \tau) = I_\alpha$. This proves I_α is a finitely generated ideal in R_α . $\square$

Proposition 1.9. *Let $f : R \rightarrow S$ be a homomorphism of commutative rings. Let $\theta : R[x_1, \dots, x_n] \rightarrow S$ be an onto homomorphism of R -algebras. Then S is an R -algebra of finite presentation if and only if $\ker \theta$ is a finitely generated ideal.*

Proof. First assume f is of finite presentation. The diagram of R -algebra homomorphisms

$$\begin{array}{ccc} R[x_1, \dots, x_n] & \xrightarrow{\theta} & S \\ & \swarrow \sigma \quad \searrow f & \\ & R & \end{array}$$

commutes, where σ is the natural map. Since σ is locally of finite type, by Proposition 1.7(5), θ is locally of finite presentation. By Lemma 1.8, $\ker \theta$ is locally a finitely generated ideal in $R[x_1, \dots, x_n]$. By Lemma 1.2, $\ker \theta$ is a finitely generated ideal. The converse is immediate. $\square$

Proposition 1.10. *If $f : R \rightarrow S$ is locally of finite presentation, then f is of finite presentation.*

Proof. There is an onto R -algebra homomorphism $\theta : R[x_1, \dots, x_n] \rightarrow S$. Let $\sigma : R \rightarrow R[x_1, \dots, x_n]$ be the natural map. By Proposition 1.7(5), θ is locally of finite presentation. Proceed as in the proof of Proposition 1.9. $\square$

Corollary 1.11. *Let $f : R \rightarrow S$ be a homomorphism of commutative rings. If $f^\# : \operatorname{Spec} S \rightarrow \operatorname{Spec} R$ is an open immersion, then S is an R -algebra of finite presentation.*

Proof. By Proposition 1.7(1), $f : R \rightarrow S$ is locally of finite presentation. By Proposition 1.10, $f : R \rightarrow S$ is of finite presentation. $\square$

Lemma 1.12 is a local to global lemma for a homomorphism of finite presentation which generalizes [3, Lemma 3.1.13] and [1, Lemma 2.1.5].

Lemma 1.12. *Let $f : A \rightarrow B$ be a homomorphism of commutative R -algebras. Assume f makes B into an A -algebra of finite presentation. If $W \subseteq R$ is a multiplicative set and $f_W : W^{-1}A \rightarrow W^{-1}B$ is an isomorphism of $W^{-1}R$ -algebras, then there exists $w \in W$ such that $f_w : A_w \rightarrow B_w$ is an isomorphism of R_w -algebras.*

Proof. As an A -algebra, B is finitely generated. Say B is generated as an A -algebra by the elements $y_1, \dots, y_m$. By [3, Lemma 3.1.11] there exists $\beta \in W$ such that $\beta(Ry_1 + \dots + Ry_m) \subseteq f(A)$. Then $f_\beta : A_\beta \rightarrow B_\beta$ is an onto homomorphism of R_β -algebras. By Lemma 1.5, f_β makes B_β into an A_β -algebra of finite presentation.

By Lemma 1.8, $(\ker f)_\beta$ is a finitely generated ideal in A_β . Say $(\ker f)_\beta = A_\beta x_1 + \cdots + A_\beta x_n$. Since $(\ker f)_\beta \otimes_R W^{-1}R = \ker f \otimes_R W^{-1}R = (0)$, there exists $\alpha \in W$ such that $\alpha(Rx_1 + \cdots + Rx_n) = (0)$. Hence $\alpha(\ker f)_\beta = (0)$. If we set $w = \alpha\beta$, then $(A_\beta)_\alpha = A_w$, $(B_\beta)_\alpha = B_w$ and $f_w : A_w \rightarrow B_w$ is an isomorphism of R_w -algebras. $\square$

2. STANDARD ÉTALE ALGEBRAS

If S is an étale R -algebra, we show that S is locally isomorphic to a standard étale algebra. Theorem 2.1, which is a generalization of [1, Theorem 9.3.9], does not require the ground ring R to be noetherian. As an application, in Theorem 2.3 below we show that $\text{Spec } S \rightarrow \text{Spec } R$ is an open map.

Theorem 2.1. *Let R be a commutative ring, S a commutative R -algebra, $\mathfrak{q}$ a prime ideal in S , and $\mathfrak{p} = \mathfrak{q} \cap R$. If S is étale over R in a neighborhood of $\mathfrak{q}$, then there exist $f \in S - \mathfrak{q}$ and $h \in R - \mathfrak{p}$ such that $S[f^{-1}]$ is isomorphic to a standard étale $R[h^{-1}]$ -algebra of the form $(R[h^{-1}][x]/(P)) [g^{-1}]$ where P is a monic polynomial in $R[h^{-1}][x]$ and the derivative P' is invertible in $(R[h^{-1}][x]/(P)) [g^{-1}]$.*

Proof. The proof is divided into four steps. As mentioned in [1, Remark 9.3.10], in the proof of [1, Theorem 9.3.9], the noetherian hypothesis on R was used for the local to global argument in Step 2. The noetherian hypothesis was also used in the last line of Step 4 to assume the kernel of h is a finitely generated ideal. For sake of completeness, we include all four steps.

Step 1: Reduction to the case where S is a finitely generated R -module. We assume S is R -separable and an R algebra of finite presentation. By [1, Exercise 9.3.5], S is quasi-finite over R . By Zariski's Main Theorem, [3, Corollary 7.4.16], there exists a factorization $R \rightarrow T \rightarrow S$ such that T is a finitely generated R -module and there exists $f \in T - \mathfrak{q} \cap T$ such that $T[f^{-1}] = S[f^{-1}]$. The reader should verify that T is étale over R in a neighborhood of $\mathfrak{q} \cap T$, and that if the theorem is true for T , it is true for S .

Step 2: Reduction to the case where R is local. We assume that S is a separable flat R -algebra of finite presentation. Assume the theorem is true for $R_{\mathfrak{p}}$. So there exist polynomials $f, g \in R_{\mathfrak{p}}[x]$ such that f is monic, f' is invertible in $(R_{\mathfrak{p}}[x]/(f)) [g^{-1}]$, $c \in S_{\mathfrak{p}} - \mathfrak{q}S_{\mathfrak{p}}$, and there is an $R_{\mathfrak{p}}$ -algebra isomorphism

$$\left(\frac{R_{\mathfrak{p}}[x]}{(f)} \right) [g^{-1}] \xrightarrow{\phi} S_{\mathfrak{p}}[c^{-1}].$$

The reader should verify that there is an element $\alpha \in R - \mathfrak{p}$ such that f and g come from $R[\alpha^{-1}][x]$, f' is invertible in $(R[\alpha^{-1}][x]/(f)) [g^{-1}]$, c comes from $S[\alpha^{-1}]$, $\phi(x)$ comes from $S[\alpha^{-1}][c^{-1}]$, and $\phi(g^{-1})$ comes from a unit in $S[\alpha^{-1}][c^{-1}]$. Therefore,

the diagram

$$\begin{array}{ccc}
 R & \xrightarrow{\theta} & S \\
 \downarrow r & & \downarrow s \\
 \left(\frac{R[\alpha^{-1}][x]}{(f)} \right) [g^{-1}] & \xrightarrow{\psi} & S[\alpha^{-1}][c^{-1}] \\
 \downarrow & & \downarrow \\
 \left(\frac{R_{\mathfrak{p}}[x]}{(f)} \right) [g^{-1}] & \xrightarrow{\phi} & S_{\mathfrak{p}}[c^{-1}]
 \end{array}$$

commutes. The vertical maps r and s are étale and the map ψ is an $R[\alpha^{-1}]$ -algebra homomorphism. The $R_{\mathfrak{p}}$ -algebra isomorphism ϕ is induced by tensoring ψ with $R_{\mathfrak{p}}$. By [3, Proposition 10.5.12 (4)], ψ is of finite presentation. We can pass from the local to the global, by [3, Proposition 10.5.15 and Lemma 10.5.16]. That is, there exists an element $\beta \in R - \mathfrak{p}$ such that

$$\left(\frac{R[\beta^{-1}][x]}{(f)} \right) [g^{-1}] \xrightarrow{\phi} S[\beta^{-1}][c^{-1}]$$

is an isomorphism of $R[\beta^{-1}]$ -algebras. So if the theorem is true for $R_{\mathfrak{p}}$, it is true for R .

Step 3: Reduction to the case where $S = R[x]$ is a cyclic extension. By the previous steps, we assume R is local, S is separable over R , and finitely generated as an R -module. By [1, Theorem 8.3.7], $\mathfrak{p}$ generates the maximal ideal of $S_{\mathfrak{q}}$, and the extension of residue fields $k_{\mathfrak{p}} \rightarrow k_{\mathfrak{q}}$ is separable. Let $\bar{S} = S \otimes_R k_{\mathfrak{p}}$. Then $\bar{S}$ is artinian and the prime ideals of $\bar{S}$ correspond to those primes of S lying over $\mathfrak{p}$. Let $\bar{\mathfrak{q}}$ be the maximal ideal of $\bar{S}$ corresponding to $\mathfrak{q}$. Then $(\bar{S})_{\bar{\mathfrak{q}}} = k_{\mathfrak{q}}$. By the Primitive Element Theorem, $k_{\mathfrak{q}} = k_{\mathfrak{p}}(t)$ is a simple extension. By [3, Theorem 4.4.13], there is $x \in S$ which maps to t in $(\bar{S})_{\bar{\mathfrak{q}}}$, and such that x restricts to 0 for all other prime ideals of $\bar{S}$. Let $C = R[x]$, and $\mathfrak{r} = \mathfrak{q} \cap C$. Then S is a finitely generated C -module and $\mathfrak{q}$ is the only prime ideal of S lying over $\mathfrak{r}$. Thus $S_{\mathfrak{r}} = S_{\mathfrak{q}}$ is a finitely generated $C_{\mathfrak{r}}$ -module. The composite map

$$k_{\mathfrak{p}}(t) \rightarrow C_{\mathfrak{r}}/\mathfrak{r}C_{\mathfrak{r}} \rightarrow S_{\mathfrak{q}}/\mathfrak{q}S_{\mathfrak{q}} = k_{\mathfrak{q}}$$

is onto, hence $C_{\mathfrak{r}} \rightarrow S_{\mathfrak{q}}$ is onto ([3, Exercise 3.2.4]). Since $C \rightarrow S$ is one-to-one, by [3, Lemma 3.1.11], there is $f \in C - \mathfrak{r}$ such that $C[f^{-1}] = S[f^{-1}]$.

Step 4: Assume R is local, $S = R[x]$ is finitely generated as an R -module, $\mathfrak{q} \in \text{Spec } S$ is the only prime lying above the maximal ideal $\mathfrak{p} \in \text{Spec } R$, and S is separable, flat and finitely presented over R . So $S/\mathfrak{p}S$ is an artinian local ring, and is separable over the field $k_{\mathfrak{p}}$. It follows from [3, Corollary 8.5.6] that $S/\mathfrak{p}S$ is equal to the field $k_{\mathfrak{q}}$. Let n be the dimension of $k_{\mathfrak{q}}$ over $k_{\mathfrak{p}}$. Then $1, x, \dots, x^{n-1}$ generate $S/\mathfrak{p}S = k_{\mathfrak{q}}$ over $R/\mathfrak{p} = k_{\mathfrak{p}}$. By [3, Corollary 3.4.1] (Nakayama's Lemma), the R -module S is generated by $1, x, \dots, x^{n-1}$. There is a monic polynomial $P \in R[X]$ such that $h : R[X]/(P) \rightarrow S$ is onto. The natural map $\bar{h} : (R/\mathfrak{p})[X]/(P) \rightarrow S/\mathfrak{p}S$ is an isomorphism. The image of P in $k_{\mathfrak{p}}[X]$ is therefore the irreducible polynomial for t over $k_{\mathfrak{p}}$, where t is as in Step 3. Since $k_{\mathfrak{q}}$ is separable over $k_{\mathfrak{p}}$, (P, P') is the unit ideal in $k_{\mathfrak{p}}[X]$. Let $\mathfrak{q}' = h^{-1}(\mathfrak{q})$. By [1, Theorem 8.1.24], $(R[X]/(P))_{\mathfrak{q}'}$ is separable over R . By [3, Proposition 8.6.1], (P, P') is the unit ideal in $(R[X]/(P))_{\mathfrak{q}'}$. By [3, Lemma 3.1.11], there is some $b \in (R[X]/(P)) - \mathfrak{q}'$ such that $(R[X]/(P))[b^{-1}]$ is a

standard étale R -algebra. If we set $c = h(b)$, then $h : (R[X]/(P))[b^{-1}] \rightarrow S[c^{-1}]$ is onto. By [1, Corollary 9.2.6], $h : (R[X]/(P))[b^{-1}] \rightarrow S[c^{-1}]$ is étale. The kernel of h is finitely generated, by [3, Lemma 10.5.13]. By [3, Exercise 3.2.4], h is a localization. $\square$

Lemma 2.2 and its proof are based on [5, Chapitre V, lemma, p. 57].

Lemma 2.2. *Let S be an R -algebra such that S is an R -progenerator module of constant rank r . If $\theta : R \rightarrow S$ is the structure homomorphism, then the continuous map $\theta^\# : \text{Spec } S \rightarrow \text{Spec } R$ is open.*

Proof. Let $\alpha \in S - (0)$. We show that $\theta^\#(U(\alpha))$ is an open subset of $\text{Spec } R$. By [1, Example 5.3.3], the characteristic polynomial of α over R exists. Let $\text{char. poly}_R(\alpha) = x^r + a_{r-1}x^{r-1} + \cdots + a_1x + a_0$ in $R[x]$. Let $\mathfrak{q} \in U(\alpha)$. Then $\alpha \notin \mathfrak{q}$. Consider $\mathfrak{p} = \mathfrak{q} \cap R$. Let $k(\mathfrak{p}) = R_{\mathfrak{p}}/\mathfrak{p}R_{\mathfrak{p}}$ be the residue field and $S \otimes_R k(\mathfrak{p})$ the fiber over $\mathfrak{p}$. Then $\mathfrak{q}$ is the image of the prime $\mathfrak{q} \otimes_R k(\mathfrak{p})$ under the map $\epsilon^\# : \text{Spec } S \otimes_R k(\mathfrak{p}) \rightarrow \text{Spec } S$. Since $\alpha \otimes 1 \notin \mathfrak{q} \otimes_R k(\mathfrak{p})$, this implies $\alpha \otimes 1$ is not in the nil radical $\text{Rad}_{S \otimes_R k(\mathfrak{p})}(0)$. Therefore, the characteristic polynomial $\text{char. poly}_{k(\mathfrak{p})}(\alpha)$ is not equal to x^r . In other words, for some i , $a_i \notin \mathfrak{p}$, which implies $\mathfrak{p}$ is in the open set $\text{Spec } R - V(a_0, a_1, \dots, a_{r-1})$. Conversely, if $\mathfrak{p}$ is in $\text{Spec } R - V(a_0, a_1, \dots, a_{r-1})$, then $\text{char. poly}_{k(\mathfrak{p})}(\alpha)$ is not equal to x^r . Then $U(\alpha) \cap \text{Spec } S \otimes_R k(\mathfrak{p}) \neq \emptyset$. There is some $\mathfrak{q} \in U(\alpha)$ such that $\mathfrak{p} = \mathfrak{q} \cap R$. $\square$

Theorem 2.3 is a generalization of [1, Proposition 9.2.9] for a non-noetherian base ring R .

Theorem 2.3. *If S is an étale R -algebra with structure homomorphism $\theta : R \rightarrow S$, then $\theta^\# : \text{Spec } S \rightarrow \text{Spec } R$ is an open map.*

Proof. Let $\mathfrak{q} \in \text{Spec } S$. We need to show that there is an open neighborhood of $\mathfrak{q}$ which maps to an open neighborhood of $\mathfrak{p} = \mathfrak{q} \cap R$ in $\text{Spec } R$. By Theorem 2.1, we can assume S is a standard étale algebra of the form $(R[x]/(P))[g^{-1}]$, where P is monic of degree r in $R[x]$. By Lemma 2.2, $\text{Spec } R[x]/(P) \rightarrow \text{Spec } R$ is an open map. The open immersion $\text{Spec } (R[x]/(P))[g^{-1}] \rightarrow \text{Spec } R[x]/(P)$ is open. The proof follows. $\square$

3. THEOREMS OF PERMANENCE FOR ÉTALE ALGEBRAS

If S is an étale R -algebra, we prove that S inherits some important properties of R . For instance, in Theorem 3.6 we show that if R is reduced, then S is reduced, and if R is normal, then S is normal. The proof is based on Theorem 2.1 and the analogous theorems of permanence for an étale R -algebra of the form $R[x]/(P)$ where P is a separable polynomial. The following results are based on [5, Chapitre VII, §1 and §2, pp. 72 – 75].

Let R be a commutative ring and $f(x) = r_0 + r_1x + \cdots + r_{n-1}x^{n-1} + x^n$ a monic polynomial in $R[x]$ of degree n . Let $S = R[x]/(f(x))$. As in the proof of [1, Proposition 4.6.1], in $R[x, y]$ there is a polynomial $q(x, y)$, the divided difference, such that $f(y) - f(x) = (y - x)q(x, y)$. Expanding q as a polynomial in y , $q(x, y) = \sum_{j=0}^{n-1} q_j(x)y^j$ for $q_j(x) \in R[x]$. From the divided difference we get the derivative

$$(3.1) \quad f'(x) = q(x, x) = \sum_{j=0}^{n-1} q_j(x)x^j.$$

In $S[y]$ we have

$$(3.2) \quad f(y) = (y - x)q(x, y) = (y - x) \sum_{j=0}^{n-1} q_j(x)y^j.$$

The R -module S is free with basis $\{1, x, \dots, x^{n-1}\}$. Let π_i be the R -module projection of S onto the coefficient of x^i . Then $\{(x_i, \pi_i) \mid i = 0, \dots, n-1\}$ is a dual basis for S and the trace map is given by $T_R^S(z) = \sum_{i=0}^{n-1} \pi_i(zx^i)$.

Proposition 3.1 and its corollary are attributed to John Tate.

Proposition 3.1. *In the above context, the following are true.*

- (1) $\{(x_i, q_i(x)\pi_{n-1}) \mid i = 0, \dots, n-1\}$ is a dual basis for S .
- (2) $T_R^S(z) = f'(x)\pi_{n-1}(z)$.

Proof. First we establish some notation. Tensoring $\pi_{n-1} : S \rightarrow R$ with $R[y]$, we have the $R[y]$ -module homomorphism $\pi_{n-1} \otimes 1 : S[y] \rightarrow R[y]$. For simplicity in the following, we will simply write π_{n-1} for both maps. The R -module homomorphism $\sigma_0 : S \rightarrow R[y]$ is defined on the free basis $\{x^i \mid i = 0, \dots, n-1\}$ by the rule $\sigma_0(x^i) = y^i$. We define $\sigma : S[y] \rightarrow R[y]$ by $\sigma(\sum_{j=0}^p s_j y^j) = \sum_{j=0}^p \sigma_0(s_j) y^j$. Let $h(y)$ be an arbitrary polynomial in $S[y]$. We show

$$(3.3) \quad f(y)\pi_{n-1}(h(y)) = \sigma((y-x)h(y)).$$

Because π_{n-1} and σ are $R[y]$ -linear, it suffices to prove Eq. (3.3) when $h(y)$ is one of the basis elements $x^i \in S$. There are two cases. If $0 \leq i < n-1$, then $\pi_{n-1}(x^i) = 0$, so the left hand side is 0. The right hand side is $\sigma((y-x)x^i) = \sigma(yx^i - x^{i+1}) = \sigma_0(x^i)y - \sigma_0(x^{i+1}) = y^i y - y^{i+1} = 0$. In this case, Eq. (3.3) is true. Now when $h(y) = x^{n-1}$, the left hand side of Eq. (3.3) is $f(y)$. The right hand side is

$$\begin{aligned} \sigma((y-x)x^{n-1}) &= \sigma(yx^{n-1} + r_{n-1}x^{n-1} + \dots + r_1x + r_0) \\ &= y^n + r_{n-1}y^{n-1} + \dots + r_1y + r_0 \\ &= f(y) \end{aligned}$$

since $f(x) = 0$ in S . Therefore Eq. (3.3) holds in every case.

For $h(y) = x^i \sum_{j=0}^{n-1} q_j(x)y^j$, Eq.(3.3) becomes

$$\begin{aligned} f(y) \sum_{j=0}^{n-1} \pi_{n-1}(x^i q_j(x)) y^j &= \sigma \left((y-x)x^i \sum_{j=0}^{n-1} q_j(x)y^j \right) \\ &= \sigma(x^i f(y)) \\ &= y^i f(y) \end{aligned}$$

where we used Eq.(3.2) to get the second equation from the first. Since $f(y)$ is monic it is not a divisor of zero. Cancelling,

$$\sum_{j=0}^{n-1} \pi_{n-1}(x^i q_j(x)) y^j = y^i$$

which implies $\pi_{n-1}(x^i q_j(x)) = \delta_{ij}$ (Kronecker delta). Therefore, $q_j(x)\pi_{n-1}$ is equal to π_j . This proves (1).

The trace formula in (2) follows from:

$$T_R^S(z) = \sum_{i=0}^{n-1} \pi_i(zx^i) = \sum_{i=0}^{n-1} \pi_{n-1}(q_i(x)zx^i) = \sum_{i=0}^{n-1} q_i(x)x^i \pi_{n-1}(z) = f'(x)\pi_{n-1}(z)$$

where the last equation is from Eq.(3.1). $\square$

Corollary 3.2. *For every $z \in S$, $f'(x)z = \sum_{i=0}^{n-1} T_R^S(q_i(x)z)x^i$.*

Proof. Using the dual basis of Proposition 3.1 (1) for the R -module S , we have

$$f'(x)z = \sum_{i=0}^{n-1} \pi_{n-1}(q_i(x)f'(x)z)x^i.$$

It follows from Proposition 3.1 (2) that

$$\pi_{n-1}(q_i(x)f'(x)z) = f'(x)\pi_{n-1}(q_i(x)z) = T_R^S(q_i(x)z).$$

$\square$

Lemma 3.3. *Let R be a commutative ring, $f(x)$ a monic polynomial of degree n in $R[x]$, and $S = R[x]/(f(x))$. Then*

$$f'(x) \operatorname{Rad}_S(0) \subseteq \operatorname{Rad}_R(0)S.$$

Proof. Let $P \in \operatorname{Spec} R$ and $k(P) = R_P/(PR_P)$, the residue field at P . Consider the commutative diagram

$$\begin{array}{ccc} S & \xrightarrow{1 \otimes \eta} & S \otimes_R k(P) \\ \uparrow & & \uparrow \\ R & \xrightarrow{\eta} & k(P) \end{array}$$

As noted in [3, Definition 8.6.5], $T_{k(P)}^{S \otimes_R k(P)} = T_R^S \otimes 1$. If z is a nilpotent element in S , then $z \otimes 1$ is a nilpotent element in $S \otimes_R k(P)$. By [2, Exercise 6.2.31], the trace of $z \otimes 1$ is 0 in $k(P)$. Since $k(P)$ is equal to the quotient field of R/P , the kernel of the natural map $\eta : R \rightarrow k(P)$ is P . Therefore, $T_R^S(z)$ is in P . Since P was arbitrary, this proves $T_R^S(z) \in \operatorname{Rad}_R(0)$. By Corollary 3.2,

$$f'(x)z = \sum_{i=0}^{n-1} T_R^S(q_i(x)z)x^i \in \sum_{i=0}^{n-1} \operatorname{Rad}_R(0)x^i \subseteq \operatorname{Rad}_R(0)S$$

since $q_i(x)z \in \operatorname{Rad}_S(0)$ for each i . $\square$

Let R be a commutative ring and $f(x)$ a monic polynomial of degree n in $R[x]$. If $R[x]/(f(x))$ is a separable R -algebra, then we say $f(x)$ is a *separable polynomial*.

Theorem 3.4. *Let R be a commutative ring such that $\operatorname{Rad}_R(0) = (0)$. If $f(x)$ is a monic polynomial in $R[x]$ and $S = R[x]/(f(x))$, then $\operatorname{Rad}_{S_{f'}}(0) = (0)$. If $f(x)$ is separable, then $\operatorname{Rad}_S(0) = (0)$.*

Proof. By Lemma 3.3, $f'(x) \operatorname{Rad}_S(0) = (0)$. By [3, Exercise 3.3.28], upon localizing we have $f'(x) \operatorname{Rad}_{S_{f'}}(0) = (0)$. Since $f'(x)$ is invertible in $S_{f'}$, $\operatorname{Rad}_{S_{f'}}(0) = (0)$. If $f(x)$ is separable, then $S = S_{f'}$ ([3, Proposition 8.6.1]). $\square$

Theorem 3.5. *Let R be an integrally closed integral domain, $f(x)$ a monic polynomial in $R[x]$ of degree n , and $S = R[x]/(f(x))$. Then $S_{f'}$ is a normal ring. If $f(x)$ is separable, then S is normal.*

Proof. Let K denote the quotient field of R . Set $\Lambda = S \otimes_R K \cong K[x]/(f(x))$ and let $\bar{S}$ denote the integral closure of S in Λ . The diagram

$$\begin{array}{ccccc} S & \longrightarrow & \bar{S} & \longrightarrow & \Lambda = S \otimes_R K \\ \uparrow & & & & \uparrow \\ R & \longrightarrow & & & K \end{array}$$

commutes, every arrow is one-to-one. Consider the trace mapping $T_K^\Lambda : \Lambda \rightarrow K$. Since $\bar{S}$ is integral over R and R is integrally closed in K , if α is an element of $\bar{S}$, then by [3, Theorem 5.1.11], $T_K^\Lambda(\alpha)$ is an element of R . By Corollary 3.2 applied to $\alpha \in \Lambda$,

$$f'(x)\alpha = \sum_{i=0}^{n-1} T_K^\Lambda(q_i(x)\alpha)x^i$$

where x denotes the coset of x in $S = R[x]/(f(x))$. Because each $q_i(x)$ is in S , we have $q_i(x)\alpha \in \bar{S}$. Hence $T_K^\Lambda(q_i(x)\alpha) \in R$. This proves $f'(x)\alpha \in S$, which implies $\bar{S} \subseteq S_{f'}$. This shows $S_{f'}$ is equal to $(\bar{S})_{f'}$. By [3, Lemma 5.1.7], $(\bar{S})_{f'}$ is the integral closure of $S_{f'}$ in $S_{f'} \otimes_R K$. Since Λ is finite dimensional over K , Λ is artinian. By Theorem 3.4, the nilradical of $S_{f'}$ is trivial. Since $S_{f'} \otimes_R K$ is a localization of $S_{f'}$, the ring $S_{f'} \otimes_R K$ is semisimple. Since $S_{f'}$ is integrally closed in the semisimple ring $S_{f'} \otimes_R K$, by [3, Lemma 11.1.6], $S_{f'}$ is normal. $\square$

Theorem 3.6 is a generalization of [1, Theorem 9.3.12] for non-noetherian rings.

Theorem 3.6. *Let S be an étale R -algebra.*

- (1) *For every $\mathfrak{q} \in \text{Spec } S$, if $\mathfrak{p} = \mathfrak{q} \cap R$, then $\dim(S_{\mathfrak{q}}) = \dim(R_{\mathfrak{p}})$.*
- (2) *If R is regular, then S is regular.*
- (3) *If R is a normal ring, then S is a normal ring.*
- (4) *If $\text{Rad}_R(0) = (0)$, then $\text{Rad}_S(0) = (0)$. In other words, if R is reduced, then S is reduced.*

Proof. Let $\mathfrak{q} \in \text{Spec } S$ and $\mathfrak{p} = \mathfrak{q} \cap R$. By Theorem 2.1, there exist $\beta \in S - \mathfrak{q}$ and $\alpha \in R - \mathfrak{p}$ such that S_β is isomorphic to a standard étale R_α -algebra of the form $(R_\alpha[x]/(P))[g^{-1}]$ where P is a monic polynomial in $R_\alpha[x]$ and P' is invertible in $(R_\alpha[x]/(P))[g^{-1}]$.

(4): Since $\text{Rad}_R(0) = (0)$, it follows that $\text{Rad}_{R_\alpha}(0) = (0)$. By Theorem 3.4, $\text{Rad}_{S_\beta}(0) = (0)$. Since $\mathfrak{q}$ was arbitrary, there is an open cover of $\text{Spec } S$ by basic open sets of the form $U(\beta)$, where S_β is reduced. From this it follows that S is reduced.

(3): Since R is normal, it follows that R_α is normal. By Theorem 3.5, S_β is normal. Therefore, $S_{\mathfrak{q}}$ is an integrally closed integral domain. Since $\mathfrak{q}$ was arbitrary, S is normal.

(1): By [3, Exercise 8.6.24], S is a quasi-finite R -algebra. By Zariski's Main Theorem, [3, Corollary 7.4.16], there exists a factorization $R \rightarrow T \rightarrow S$ such that T is finitely generated as an R -module and $\text{Spec } S \rightarrow \text{Spec } T$ is an open immersion. Let $\mathfrak{r} = \mathfrak{q} \cap T$. Then $T_{\mathfrak{r}} \cong S_{\mathfrak{q}}$. It suffices to show $\dim R_{\mathfrak{p}} = \dim T_{\mathfrak{r}}$. Let $T_{\mathfrak{p}} = T \otimes_R R_{\mathfrak{p}}$. Then $T_{\mathfrak{p}}$ is finitely generated as an $R_{\mathfrak{p}}$ -module. Since $T_{\mathfrak{p}}$ is a flat $R_{\mathfrak{p}}$ -algebra and

the prime ideal $\mathfrak{r} \otimes_R R_{\mathfrak{p}}$ in $T_{\mathfrak{p}}$ lies over the maximal ideal $\mathfrak{p}R_{\mathfrak{p}}$, by [3, Lemma 3.5.5], $T_{\mathfrak{p}}$ is faithfully flat over $R_{\mathfrak{p}}$. It follows from [3, Theorem 5.3.6 (3)] that a chain $Q_0 \subsetneq Q_1 \subsetneq \cdots \subsetneq Q_n$ of length n in $\text{Spec}(T_{\mathfrak{p}})$ gives rise to a chain $Q_0 \cap R_{\mathfrak{p}} \subsetneq Q_1 \cap R_{\mathfrak{p}} \subsetneq \cdots \subsetneq Q_n \cap R_{\mathfrak{p}}$ of length n in $\text{Spec}(R_{\mathfrak{p}})$. Thus $\dim(T_{\mathfrak{p}}) \leq \dim(R_{\mathfrak{p}})$. To finish the proof, it suffices to show that the height of $\mathfrak{r} \otimes_R R_{\mathfrak{p}}$ is greater than or equal to $\dim R_{\mathfrak{p}}$. By [3, Proposition 5.3.4] applied iteratively, a chain $P_0 \subsetneq P_1 \subsetneq \cdots \subsetneq \mathfrak{p}R_{\mathfrak{p}}$ of length n in $\text{Spec}(R_{\mathfrak{p}})$ lifts to a chain $Q_0 \subsetneq Q_1 \subsetneq \cdots \subsetneq \mathfrak{r} \otimes_R R_{\mathfrak{p}}$ of length n in $\text{Spec}(T_{\mathfrak{p}})$.

(2): By hypothesis, $R_{\mathfrak{p}}$ is a noetherian regular local ring. Say $\dim(R_{\mathfrak{p}}) = d$. Then $S_{\mathfrak{q}}$ is a finitely generated $R_{\mathfrak{p}}$ -algebra, hence is noetherian, by the Hilbert Basis Theorem. By (1), $\dim(S_{\mathfrak{q}}) = d$. The maximal ideal of $R_{\mathfrak{p}}$ is generated by a regular system of parameters $\{\pi_1, \dots, \pi_d\}$. Since $S_{\mathfrak{q}}$ is separable over $R_{\mathfrak{p}}$, $\{\pi_1, \dots, \pi_d\}$ is a generating set for the maximal ideal of $S_{\mathfrak{q}}$ ([3, Exercise 8.5.9]). This shows $S_{\mathfrak{q}}$ is a regular local ring. $\square$

REFERENCES

- [1] Timothy J. Ford, *Separable algebras*, Graduate Studies in Mathematics, vol. 183, American Mathematical Society, Providence, RI, 2017. MR 3618889
- [2] ———, *Introduction to abstract algebra*, Open Math Notes, American Mathematical Society, Reference # OMN:202006.110827, 2024, available at the URL: <https://www.ams.org/open-math-notes/omn-view-listing?listingId=110827>.
- [3] ———, *Commutative algebra*, Open Math Notes, American Mathematical Society, Reference # OMN:202409.111443, 2024, the latest version is available at the URL: <https://tim4datfau.github.io/Timothy-Ford-at-FAU/preprints/CA.pdf>.
- [4] A. Grothendieck, *Éléments de géométrie algébrique. IV. Étude locale des schémas et des morphismes de schémas. I*, Inst. Hautes Études Sci. Publ. Math. (1964), no. 20, 259. MR 0173675 (30 #3885)
- [5] Michel Raynaud, *Anneaux locaux henséliens*, Lecture Notes in Mathematics, Vol. 169, Springer-Verlag, Berlin-New York, 1970. MR 0277519 (43 #3252)

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